# RAGstofan

Asking Statistics Iceland in plain English or Icelandic

Gagnaþon 2026 · Track 2: an agentic AI over Statistics Iceland's PX-API

## What we set out to do

Statistics Iceland (Hagstofa Íslands) publishes 1,857 tables through its PX-API. The numbers are there, but finding them means knowing which table holds them, which of its dimensions and categories to choose, and often doing a calculation on top. RAGstofan lets anyone ask in plain English or Icelandic ("*Hve mörgum fleiri karlar en konur búa á Íslandi?*") and get back the official figure, its unit, the table it came from with a link, a chart, and every step that produced it.

## Our approach: a strong harness around the model

We did not rely on a large model knowing the answer. We built a **harness**: a pipeline in which the language model makes small, checkable decisions, each with a fixed output schema, and code does everything that can be checked or computed. The model reads, plans and chooses; it never does arithmetic and never has the last word on a number. Code checks each of its choices against the table, does every calculation in Python, and compares every number in the written answer with the data before it is shown. The harness runs on open models through any OpenAI-compatible endpoint (Qwen models through Ollama on a GB10 GPU server, "the Spark"; the same code runs on a laptop), and every answer shows its source table, categories, calculation and steps, so a user can check it.

**Our main focus was finding the correct table.** Once the right table is found, reading the right number from it is a much smaller problem; with the wrong table, no model can give the right answer. Most of our work, and most of our measuring, went into the table search.

## How we find the table

**The dictionary.** A large share of our effort went into a bilingual dictionary (`data/keywords.json`) that maps the words people actually use to Statistics Iceland's subjects and tables. It holds 212 subjects with 2,810 English and Icelandic words and inflections (*verðbólga, verðbólgan, inflation, cost of living* → consumer price index → VIS01000), and for 171 of them the tables that answer it, checked against the catalogue's metadata. It closes the gap between everyday language and official titles, which keyword and vector search cannot close on their own.

**Three methods, each confirmed.** The whole question is first treated as one part, to be answered from one table. Three methods each suggest a table, and the model answers yes or no: does this table hold the figures asked for?

1. the **dictionary**: the table of the most specific subject the question names;
2. **search** over all tables: BM25 keyword search (over titles, dimensions and category labels, in both languages) and multilingual e5 vectors run in parallel and are fused by rank; code rules then drop variants nobody asked for (seasonally adjusted, forecasts, old index bases, tables without the place or year asked);
3. **every table**: the model names its best three from a list of all tables, sized to its context window.

If no table is confirmed, the question is split into 2, 3 and then 4 parts, each searched the same way; if a part still has none, the user picks from each method's top three.

**After the table.** The model chooses the categories (year, place, sex, ...); code checks each choice and applies the user's own words (only the places named, the period asked, quantity or value). Code does the arithmetic (differences, changes, shares, sums, rankings, figures per inhabitant across two tables), picks the chart, and checks the written answer's numbers. Figure 1 shows the whole path.

**Connection to Statistics Iceland.** A crawler walked the PX-API's table tree once and stored every table's ID, titles in both languages, dimensions, periods and category labels (the catalogue that search runs over). For a chosen table the app sends a `GET` for its metadata and a `POST` asking only for the categories needed, in json-stat2 format, to `https://px.hagstofa.is/pxen/api/v1/en/<path>/<table>.px`. Calls are spaced at least one second apart, retried on `429`, and cached. By default the same documents are read from the organisers' snapshot of every table (18 October 2026), and the live API is asked only for tables or periods the snapshot lacks.

**Interfaces.** A website with charts and voice input, an HTTP API including the Track 2 endpoint `/spyrja_px_api`, an iPhone and iPad app, and an MCP server so Claude, ChatGPT and other agents can call it.

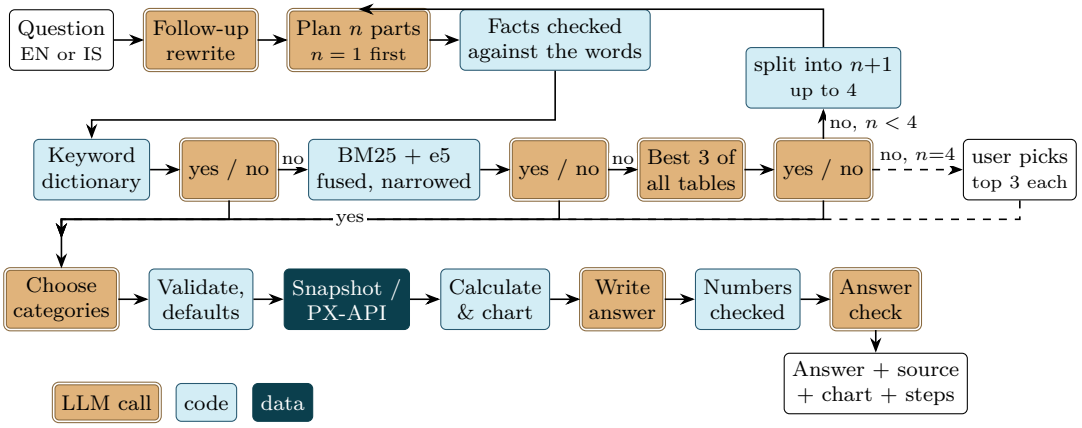



Figure 1: RAGstofan's pipeline. Each part of a question goes through the table finder (middle row) in parallel; a yes from any method, or the user's pick, leads to reading the numbers (bottom row). Python does all arithmetic.

## Test sets we built

Beyond the organisers' 60 questions, we built our own question sets to test table finding at scale, without writing answer keys by hand:

- **Questions written from the tables.** A language model read each sampled table's titles and dimensions from the catalogue and wrote three questions about it, in English and Icelandic: a plain lookup, a change between two periods, and a question about one of its categories. The table it was written from is the right answer. It also reworded each organiser question three ways (shorter, colloquial, with a typo), and wrote questions that only one of two near-identical tables can answer (base year, monthly vs yearly, old vs new series), the confusion behind most of our misses. In all about 1,300 questions over 141 tables, split by table into a half we fix against and a frozen half we only measure on.
- **Questions from Statistics Iceland's news.** We read the newest news releases on Statistics Iceland's website, and a language model wrote questions and answers from each, in English and Icelandic. A question was kept only if its answer appears word for word in the article, so the answer key is Statistics Iceland's own. These 108 questions use real topics and real phrasing, and are kept for measuring only.

## Results

**Benchmark.** We measured against the organisers' 60 questions over Statistics Iceland's data (release of 18 October 2026): 20 easy, 20 medium and 20 hard, each asked in English and Icelandic, with official answers. An answer counts as right only if its number matches the official one; "right table" means the source is among the accepted tables. The questions range from single lookups (population on 1 January) to comparisons, changes, shares and figures per inhabitant across two tables.

| Organisers' 60 questions, Qwen 3.6 35B-A3B (1 Nov 2026) | Right answer | Right table | Median time | LLM calls (median) |
|---|---|---|---|---|
| All questions | **59/60** | 59/60 | 7.4 s | 5 |
| easy / medium / hard | 20/20 / 20/20 / 19/20 | | | |
| English / Icelandic | 30/30 / 29/30 | | | |
| Same questions, Qwen 3.8 27B | **60/60** | | | |
| Latest run, 35B-A3B (one server error) | 57/60 | 58/60 | 7.3 s | 5 |
| Team's own sets, 35B-A3B: core / topics / friends | 10/10 / 20/20 / 10/10 | | | |

The one miss in the best run is a hard Icelandic question that asks for overnight stays per inhabitant: the answer gave the population but not the division. In the latest run the misses were a server error, a comparison of two municipalities answered without the difference, and the same per-inhabitant question. The 90th percentile time was 11 s and the slowest answer 21 s.

**What moved the score.** The first version, with a 7B model and keyword search over titles only, answered 1 of our 10 development questions. Search over category labels, prefix matching, synonyms and recency brought the right table into the candidates for 9 of 10, but a 14B model still answered only 3 or 4. The largest gains came from turning table-choice rules into code (drop seasonally adjusted, forecast and census tables unless asked; a named place must be an option; with no period asked, use the latest edition), which took the same model to 10/10. The

same rules written into the prompt were followed unreliably. On 20 unseen questions the first score was 11/20; after fixing what they exposed, 20/20.

**Search alone.** Without any model, on 90 questions (our 30 plus the organisers' 60), the right table was among the top 15 search hits for 74 with BM25, 62 with e5 and 78 with the two fused; e5 found the right table for 6 questions where BM25 did not. The model's checks then choose among these.

## Limits and next steps

The table finder and question splitting were redesigned after the runs above (one table first, a yes/no per method, splitting only when needed) and have not yet been scored on the 60 questions; that run is next. Each part of a question must be answerable from one table, with at most four parts. Numbers are as recent as the snapshot, with newer periods from the live API. Icelandic answers are worded less well than English ones on smaller models, though their numbers are checked the same way. RAGstofan does not forecast, give opinions or explain causes.